DSC

Donor Reactivation Campaign

Predictive Model for Targeted Outreach

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Agenda

1. The Challenge & Our Approach
2. Data & Features Used for Predictions
3. Donor Profile: Who Are Our Best Prospects?
4. Model Performance: Lift, Cumulative Gains & AUC
5. Business Case: ROI & Recommendation

The Challenge

Random Approach (Contact Everyone)

- Contact **25,636 donors** → Only 117 respond
- Response rate: **0.46%**
- Cost: **€20,509** | Revenue: €7,020
- Result: **€13,489 LOSS**

Our Solution: Predictive Targeting

- Use ML to identify high-potential donors
- Contact only the **top 7%** most likely to donate
- Focus on donations of **€60 or more**
- Transform loss into **profit**

What Information Powers Our Predictions?

Demographic data

1. Gender
2. Age and **Age group**
3. Province and Region
4. Language Preference

Giving Behavior (RFM)

- R — Recency
- F — Frequency: Number of donations
- M — Monetary: **Total**, avg, max amount
- \+ **Has donated €60+ before**

Engagement patterns

- --> Length of relationship (LOR)
- --> Donation trend over time
- --> Spontaneous donation rate
- --> Campaign response history

Key predictors

Our XGBoost model leverages **all 42 features** to capture complex patterns. The most influential are:

- Demographics: **Gender** and **Senior status** (~21%)
- RFM: Frequency, Recency, Total giving (~12%)
- Monetary: Min, Avg, **Max donation amounts** (~20%)

Who Are Our Best Prospects?

🎯Ideal Donor Profile

Age Group	**Senior (57+ years avg)**
Gender	**Slightly more female**
Previous €60+ donations	**9.2% (vs 4.5% general)**
Average past donation	**€25.22**
Maximum past donation	**€27.90**
Total lifetime giving	**€67.57**

Top 7% vs General Population

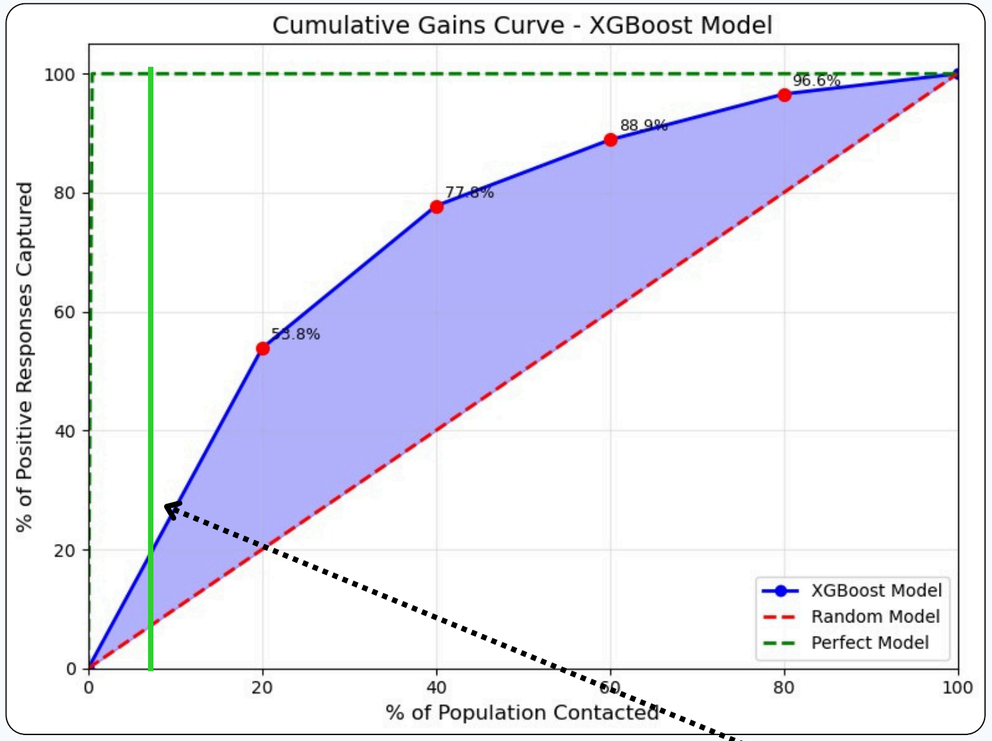
Senior Status	**33.8% vs 18.4%**
Has €60+ Donation History	**9.2% vs 4.5%**
Donation Frequency	**1.38 vs 2.66 donations**

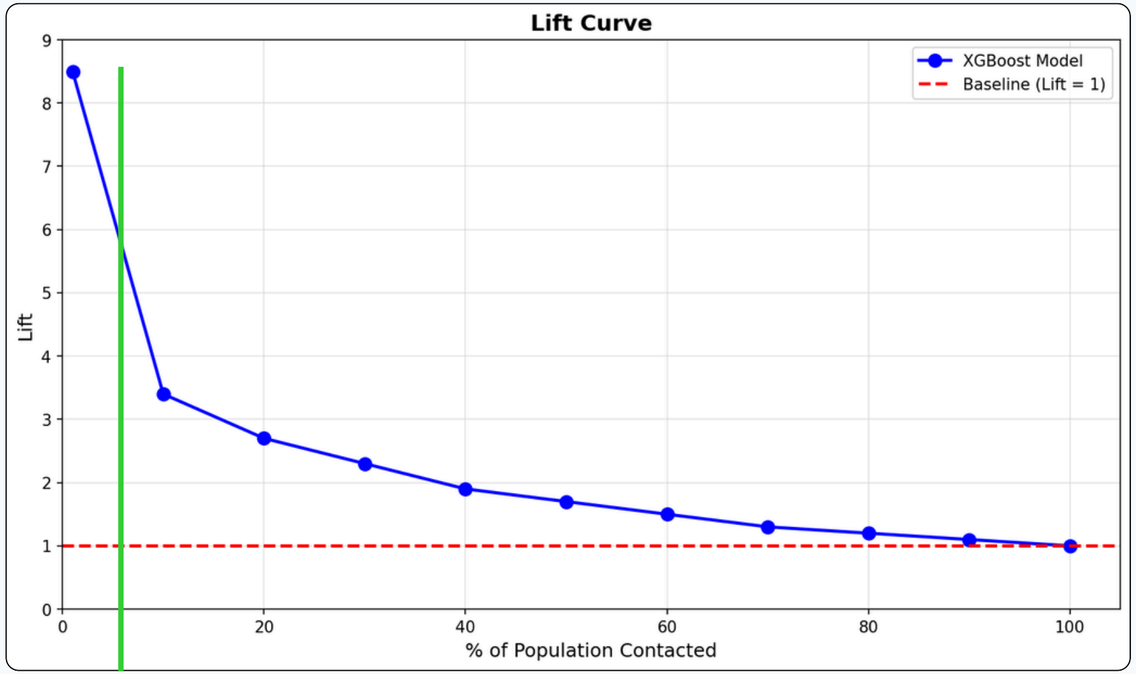
💡Actionable Insight

Prioritize **senior donors** with a proven **€60+ giving capacity** and consistent donation history. They show the highest likelihood to respond.

Model Performance: XGBoost

0.76	4.4x	30.8%	2.01%
AUC score	Lift at 7%	Cumulative Gain	Response Rate





7% = optimal threshold

Business Case: The Value of Our Model

Metric	Random	With Model
Letters sent	25,636	**1,794**
Campaign Cost	€20,509	**€1,435**
Donors Reached	117	**36**
Response rate	0.46%	**2.01%**
Revenue	€7,020	**€2,160**
Profit	**-€13,489**	**+€725**

€14,214

Total Value Created

93%
Cost reduction

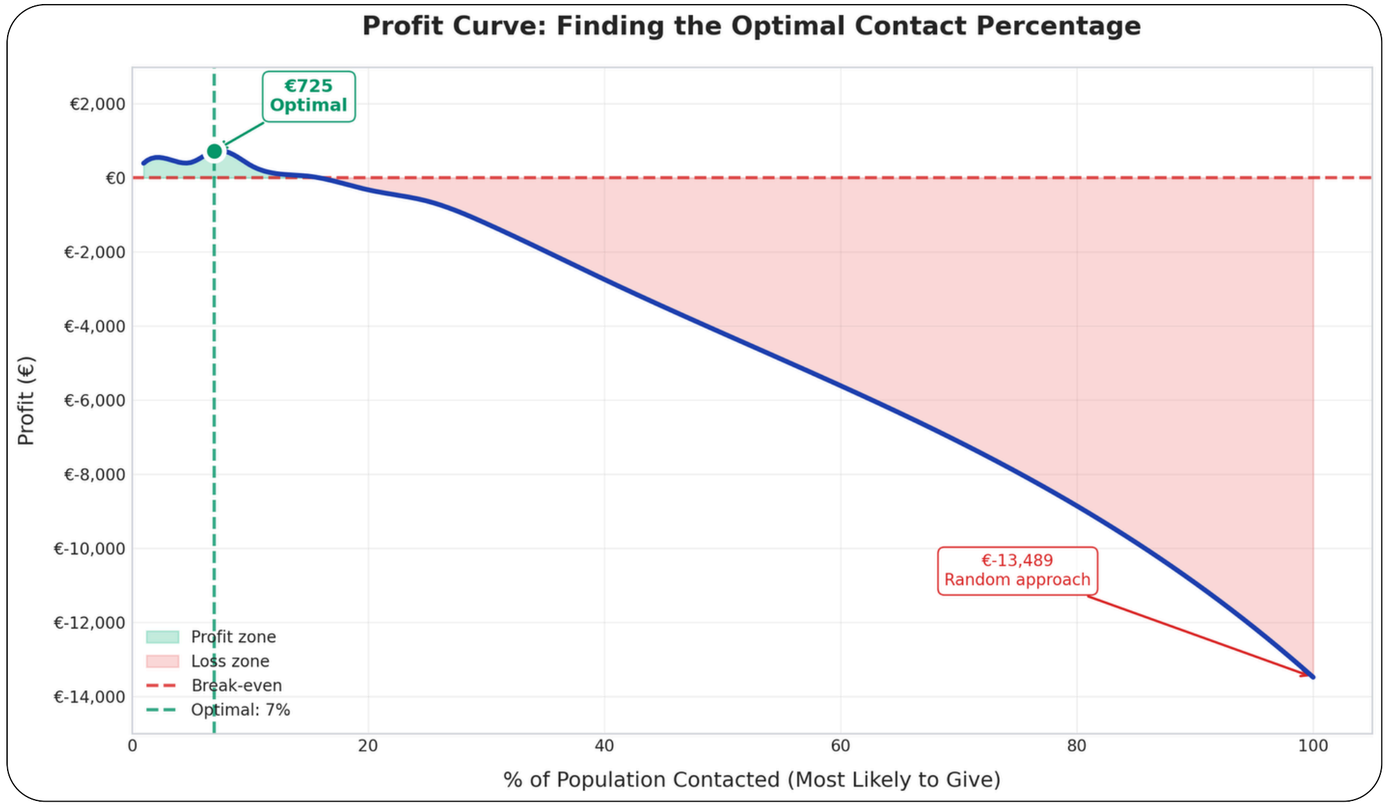
4.4x
Response lift

Key Takeaway

The model transforms a €13,489 loss into a €725 profit by reducing mailing costs by 93% while still capturing 31% of potential donors.

Business Case: The Value of Our Model

% POP MOST LIKELY	LETTERS	COST	DONORS	CUMUL. GAIN	RESP. RATE	LIFT	PROFIT
1.0%	256.0	€205	10.0	8.5%	3.91%	8.6x	€395
2.0%	512.0	€410	16.0	13.7%	3.12%	6.8x	€550
5.0%	1,281.0	€1,025	24.0	20.5%	1.87%	4.1x	€415
7.0%	1,794.0	€1,435	36.0	30.8%	2.01%	4.4x	€725
9.0%	2,307.0	€1,846	40.0	34.2%	1.73%	3.8x	€554
10.0%	2,563.0	€2,050	40.0	34.2%	1.56%	3.4x	€350
15.0%	3,845.0	€3,076	52.0	44.4%	1.35%	3.0x	€44
20.0%	5,127.0	€4,102	63.0	53.8%	1.23%	2.7x	-€322
25.0%	6,409.0	€5,127	75.0	65.0%	1.17%	2.6x	-€627
30.0%	7,690.0	€6,152	82.0	70.1%	1.07%	2.3x	-€1,232
40.0%	10,254.0	€8,203	91.0	77.8%	0.89%	1.9x	-€2,743
50.0%	12,818.0	€10,254	101.0	86.3%	0.79%	1.7x	-€4,194
Random	**25,636.0**	€20,509	117.0	100.0%	0.46%	1.0x	-€13,489



Our Recommendation

1,794	€1,435.2	€724.8	50.51%
Donors to Contact	Campaign Budget	Expected Profit	ROI

Use the XGBoost model to target the top 7% most likely donors.

Focus on senior donors with proven €60+ giving capacity.

Thank You

If you have any question, feel free to ask